Approximating Euler's number

1 Use a calculator or other technology to approximate the following values correct to four decimal places:

a e^4 **b** e^{-1} **c** $e^{\frac{1}{5}}$ **d** $5\sqrt{e}$

e $\dfrac{4}{e}$ **f** $\dfrac{8}{9e^4}$

2 The natural base e, Euler's number, is defined as:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

The table shows the values of $\left(1 + \dfrac{1}{n}\right)^n$ for certain values of n:

n	$\left(1 + \dfrac{1}{n}\right)^n$
1	2
100	$1.01^{100} = 2.704\,813...$
1000	$1.001^{1000} = 2.716\,923...$
$10\,000$	$1.0001^{10000} = 2.718\,145...$
$100\,000$	$1.000\,01^{100000} = 2.718\,268...$

a Evaluate $\left(1 + \dfrac{1}{n}\right)^n$ for $n = 1\,000\,000$, correct to six decimal places.

b Find a decimal approximation of e correct to nine decimal places.

3 It is possible to approximate e^x using the following formula:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots + \frac{x^n}{n!} + \ldots$$

The more terms we use from the formula, the closer our approximation becomes.

a Use the first five terms of the formula to estimate the value of $e^{0.7}$.
Round your answer to six decimal places.

b Use the e^x key on your calculator to find the value of $e^{0.7}$.
Round your answer to six decimal places.

c What is the difference between these two results?

Exponential expressions

4 Simplify the following expressions:

a $3e^x + 4e^x$

b $2e^x + 7e^x$

c $5e^x - 2e^x$

d $-3e^x + 4e^x$

e $12e^{2x} - 5e^x + 7e^x$

f $2e^{2x} - 8e^x - 7e^{2x}$

g $e^x \times e^2$

h $e^x \times e^{4x}$

i $e^{3x} \times (e^x)^2$

j $\dfrac{(e^{3x})^4}{e^{5x}}$

k $\dfrac{4e^{2x} - 6e^x}{2e^x}$

l $\dfrac{e^{2x} + 7e^x}{6e^{-x}}$

5 Expand and simplify the following expressions:

a $e^x (e^x + 3)$

b $e^{2x} (e^x - 9)$

c $(e^x + 2)(e^x + 5)$

d $(e^x + 2)(e^x - 5)$

e $2e^{3x} (e^{-2x} + 1)$

f $3e^x (e^{-x} + e^x)$

g $(e^x + e^{-x})^2$

h $(e^x - e^{-2x})^2$

6 Fully factorise the following expressions:

a $8e^x + 12$

b $8e^{2x} + 12e^x$

c $8e^{4x} - 12e^{3x}$

d $8e^{4x} - e^{7x}$

e $e^{2x} - 5e^x + 6$

f $e^{2x} + 6e^x + 9$

g $e^{3x} + 6e^{2x} + 8e^x$

h $e^{4x} - 7e^{2x} - 10$

Exponential graphs

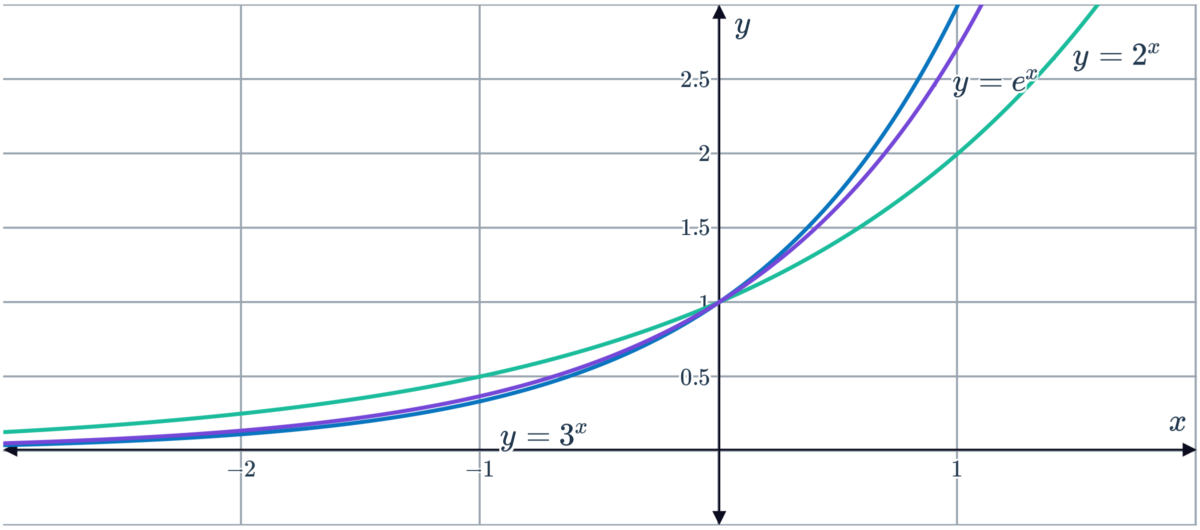
7 Consider the function $f(x) = e^x$.

a Complete the following table of values. Round each value to two decimal places.

x	-5	-3	-1	0	1	3	5
$f(x)$							

b Sketch the graph of $f(x) = e^x$ for $-5 \leq x \leq 5$.

8 The functions $y = 2^x$, $y = e^x$ and $y = 3^x$ ha[redacted]een sketched on the same axes:



a For what values of x does the inequality $3^x > e^x$ hold?

b For what values of x does the inequality $2^x > e^x$ hold?

9 Consider the function $f(x) = e^{-x}$.

a Can the function ever have a negative value?

b

i As the value of x gets larger and larger, what value does $f(x)$ approach?

ii As the value of x gets smaller and smaller, what value does $f(x)$ approach?

c Can the value of $f(x)$ ever be equal to 0?

d Find the value of $f(x)$ when $x = 0$.

e How many x-intercepts does the function have?

f Sketch the graph of $f(x) = e^{-x}$.

10 Sketch both $y = e^x$ and $y = e^{-x}$ on the same set of axes. What are the coordinates of their intersection point?

11 Consider the function $f(x) = -e^x$.

a Complete the following table of values. Round each value to three decimal places.

x	-9	-6	-3	0	3	6	9
$f(x)$							

b Are there any values of x where $f(x)$ is positive?

c Are there any values of x where $f(x)$ is equal to 0?

d Is the function increasing or decreasing?

e How would you describe the rate of increase or decrease of the function?

Transformations of exponential graphs

12 Using a graphing calculator, plot each set of three graphs below on the same screen and determine whether or not they share the same:

i y-intercept

ii Asymptote

iii Range

a $y = e^x$, $y = e^x + 2$, and $y = e^x - 3$

b $y = e^x$, $y = e^{x-2}$, and $y = e^{x+3}$

c $y = e^x$, $y = e^{2x}$, and $y = e^{3x}$

d $y = e^x$, $y = 4e^x$, and $y = -4e^x$

13 For each of the following functions, state whether they are increasing or decreasing:

a $f_1(x) = e^x$

b $f_2(x) = e^{-x}$

c $f_3(x) = -e^x$

d $f_4(x) = -e^{-x}$

14 Sketch the curves $y = e^x$, $y = e^x + 3$, and $y = e^x - 4$ on the same number plane.